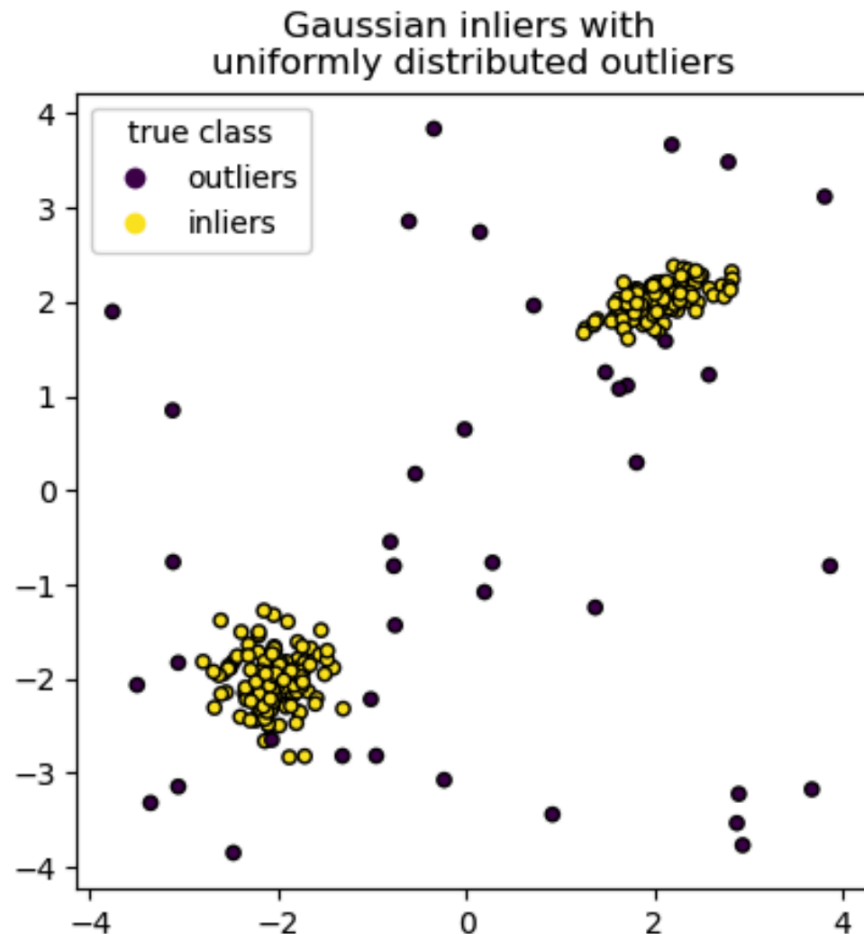


- **Examples:** Variational Autoencoders, Denoising Autoencoders
- **Applications:** Image denoising, feature extraction, anomaly detection.

Example – **Clustering:** *An Isolation Forest trained on a toy dataset*



An Isolation Forest trained on a toy dataset, where normal data forms two distinct clusters (yellow points) and anomalies are the scattered points far from those clusters (purple points). The algorithm isolates outliers by randomly partitioning data; fewer random splits are required to isolate an outlier than an inlier. In this way, the Isolation Forest flags points that stand apart from the dense clusters as anomalies for further investigation.

### 3. Reinforcement Learning

**Reinforcement learning (RL)** is a learning paradigm where an agent learns how to make decisions by interacting with an environment. There are no labeled examples of correct